# B. Mickey Mouse Constructive

time limit per test 1.5 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1100 | Tags: constructive algorithms, dp, greedy, math | https://codeforces.com/problemset/problem/2211/B

Given an array $a$, let $f(a)$ be the number of ways to partition $a$ into one or more subarrays[] such that:

- Each element appears in exactly one subarray.
- All subarrays have the same sum.

For example, if $a$=[1,1], then $f(a)$=2, because there are two such ways to partition [1,1]:

- [1,1], where the only subarray has sum 2.
- [1]+[1], where both subarrays have sum 1.

You are given two integers $x$ and $y$. Find the minimum value of $f(a)$ over all arrays $a$ of length $x+y$, consisting of $x$ copies of the number 1, and $y$ copies of the number -1 in some order. Since this answer may be large, output the answer modulo 676 767 677. Additionally, you should construct one array that achieves this minimal value.

[]An array $b$ is a subarray of an array $c$ if $b$ can be obtained from $c$ by the deletion of several (possibly, zero or all) elements from the beginning and several (possibly, zero or all) elements from the end.

## Input

Each test contains multiple test cases. The first line contains the number of test cases $t$ ($1 \le t \le 10^4$). The description of the test cases follows.

The first line of each test case contains two integers $x$ and $y$ ($0 \le x,y \le 2 \cdot 10^5$). It is guaranteed that $x+y \ge 1$.

It is guaranteed that the sum of $x$ over all test cases does not exceed $2 \cdot 10^5$, and the sum of $y$ over all test cases does not exceed $2 \cdot 10^5$.

## Output

For each test case, output two lines: the minimum value of $f(a)$ over all valid arrays $a$ modulo 676 767 677, and an example of an array that achieves the minimal result. Note you are minimizing $f(a)$, and taking that minimum value modulo 676 767 677, not finding the minimal possible result of $f(a)$ mod 676 767 677.

## Example

| Input |
| --- |
| 4<br>2 0<br>1 1<br>6 7<br>1 3 |

**Output**

```
2
1 1
1
1 -1
1
-1 1 -1 1 -1 1 -1 1 -1 1 -1 1 -1
2
-1 -1 -1 1
```

## Note

In the first test case, $x$=2 and $y$=0. The only possible array is $a$=[1,1], with $f(a)$=2 as explained above.

In the second case, $x$=1 and $y$=1. One possible array that minimizes $f(a)$ is $a$=[1,-1], where $f(a)$=1 (as the only way to partition into subarrays with all subarrays having equal sum is [[1,-1]]).